

Pokhara University
Faculty of Science and Technology

Course Code	: CMP 116	Full Marks	: 100
Course Title	: Computer Fundamentals and Applications (3-3-2)	Pass Marks	: 45
Nature of the Course	: Theory and Practical	Total Lecture Hours	: 48
Level	: Bachelor	Program	: BCA

1. Course Description

This course is designed to conceptualize the computer fundamental knowledge and its application to solve the real-life problems. This course covers the concept of computer, computer organization, software, hardware, functional behavior of peripherals and important accessories, data communication, computer networks and burning technologies. In this syllabus unit I and II covers the fundamental concept of computer; unit III and VII covers the software, unit IV and V covers the hardware where unit VI and VIII covers the data sharing tools and trends.

2. Course Objectives

The general course objectives of this course are outlined as:

1. to familiarize the student with fundamental knowledge about computer system
2. to make the student knowledgeable about software and its working procedure
3. to enhance the knowledge about various components as input, storage, processing and output devices
4. to equip the student about assembling and disassembling, general maintenance and networking the computers

3. Methods of Instruction

The medium of instruction is English and the faculty members can apply the various tools and techniques for teaching methodologies. As per the student's ability subject teachers faculties have their rights to use the teaching methodologies like Class Room/Lecture-based, Discussion-based, Project-based learning(PBL), Problem-based Learning (PBL), Flipped Classroom-based, Active Learning, Socratic Method, Cooperative Learning, Experimental Learning, Gamification, Inquiry-based Learning, Constructivist Approach, Collaborative Learning, Direct Instruction, Differentiated Instruction, Montessori Method, Reggio Emilia Approach, Waldorf Education, Peer Teaching etc. are called alternative method of teaching to motivate the students for learning. After completion of each unit, faculty members can evaluate the students theoretically and practically as well as whenever possible. They can conduct VIVA, Supersized test, Questionnaire test, Assignment test, Project work and Practical work, even Terminal examination as per the requirement.

4. Course Contents in Details

Specific Unit wise Objective	Course Contents
Unit 1: Introduction to Computer	5 hours
1) Explain the conceptual knowledge of computer with historical and generation background	1.1 Definition 1.2 History of computer 1.3 Generation of computer
2) Describe the types of computer with characteristics and applications	1.4 Types of Computer 1.5 Characteristics of computer 1.6 Applications of computer

Unit 2: Basic Organization of Computer		4 hours
1) Explain the organization of computer with their functional components 2) Describe the working mechanism of computer system	2.1 Basic function of computer 2.2 Basic functional organization of computer 2.2.1 Input unit 2.2.2 Output unit 2.2.3 Storage unit 2.2.4 Arithmetic and logic unit 2.2.5 Control unit 2.2.6 Central Processing Unit (CPU) 2.3 The system concept	
Unit 3: Computer Software		6 hours
1) Describe the concept of computer software with their roles 2) Explain the software architectures their application	3.1 Introduction to software 3.2 Logical system architecture 3.3 Types of software 3.3.1 System Software 3.3.2 Application Software 3.3.3 Utility Software 3.4 Firmware 3.6 Middleware	
Unit 4: Computer Accessories		9 hours
1) Familiarized with input and output accessories with their application 2) Identify the role of agronomical design in related devices	4.1 The Input Accessories 4.1.1 Keyboard Devices 4.1.2 Point and Draw Devices 4.1.3 Yoke 4.1.4 Data Scanning Devices 4.1.5 Digitizer 4.1.6 Microphone 4.1.7 Electronic Cards Based Devices 4.1.8 Speech Recognition Devices 4.1.9 Vision Based Devices 4.2 The Output Accessories 4.2.1 Monitor 4.2.2 Printer 4.2.3 Plotter 4.2.4 GPS 4.2.5 Projectors 4.2.6 Headphones 4.2.7 Soundcard/video Card 4.2.8 Speaker 4.2.9 Voice Response System 4.2.10 Computer Output Microfilm (COM) 4.2.11 SGD Speech Generation Device 4.3 Ergonomically Designed Devices	
Unit 5: Storage Devices		7 hours
Specific Unit wise Objective	Course Contents	
1) Describe the characters of primary and secondary storage devices 2) Explain the function and behavior of different types of memories	5.1 Introduction 5.2 Types of Storage Devices 5.3 Cache memory 5.4 Registers 5.5 Primary Memory	ROM

	5.5.1 Introduction 5.5.2 Characteristics of main memory 5.5.3 Types of primary memory 5.5.4 Difference between RAM and ROM, DRAM and SRAM, primary and secondary memory 5.6 Secondary Memory 5.6.1 Introduction 5.6.2 Characteristics of secondary memory 5.6.3 Types of Secondary memory 5.6.4 Difference between Hard disk and SSD, magnetic and optical disk, direct and sequential access magnetic memory
Unit 6: Data Communication and Computer Network	
	5 hours
1) Interpret the method of data communication including transmission modes and media 2) Explain the relationship of computer network and topologies with their advantages and disadvantages 3) Compare the role of email and internet in modern life	6.1 Data Communication 6.1.1 Introduction 6.1.2 Basic Elements of a communication system 6.1.3 Data Transmission Modes (Simplex, Half Duplex, Full Duplex) 6.1.4 Data transmission Speed (Narrowband, Voice band, Broadband) 6.1.5 Data Transmission Media (Twisted-pair wire, coaxial cable, Optical fibers, Microwave system Communication satellite) 6.2 Computer Network 6.2.1 Definition 6.2.2 Types of Computer Network (PAN, LAN, CAN, MAN and WAN), Differences, advantages disadvantages 6.2.3 Network Topologies, advantages, disadvantages 6.2.4 Intranet, Extranet, Internet 6.2.5 E-mail
Unit 7: Operating System	
	6 hours
1) Explain the features of operating system and its historical background 2) Describe the functional and architecture behaviors of various type of OS. 3) Explain the different installation methods for operating system	7.1 Introduction 7.2 History and evolution of OS 7.3 Objective of OS 7.4 Generation of OS 7.5 Functions of OS 7.6 Types of OS 7.7 System architecture of OS 7.8 Different between GUI and TUI/CUI 7.9 The booting system 7.10 Windows/ Linux operating system, settings, properties and installation guides
Unit 8: AI and Emerging Technologies	
	6 hours
1. Familiarize with Artificial Intelligence and its applications. 2. Explain about emerging trends of technologies.	8.1 Introduction to AI 8.2 AI and its Applications 8.2.1 Natural Language Processing 8.2.2 Machine vision 8.2.3 Expert system 8.3 Machine Learning 8.4 Neural Networks

5. Laboratory Work

A. Office Automation

a) Word-processing

1. Basic options of word-processing for typing, editing, formatting, margin setting, viewing, designing, printing a document.
2. Creating, inserting, editing, formulating table for word-processing
3. Document preparation with table, figure, page number, margin setting etc. and printing document as a report submission.

b) Spreadsheet

1. Preparing sheet for data processing like, arithmetic, logical and other types of functional operation
2. Preparing data table for calculation, analysis and creating various charts for presentation.
3. Inserting picture, table, graphs into word-processing
4. Printing documents after proper setting into the required format.

c) Presentation

1. Creating various types of slides with master slide for presentation
2. Setting the slide into the required format.

B. Email and Internet

1. Setting various kinds of email account and using them for personal and group purpose
2. Uploading and downloading the information from internet.

6. Prescribed Books and References

Text Books

P.K. and Priti Sinha, *Foundations of Computing*, BPB Publications, Third Edition

References Books

- a. A Text Book of Computer Fundamental and Application, OCEM Publication, (Second Edition), by Prof. Er. Hari Bhandari
- b. Computer Fundamentals, BPB Publications by V.K Jain
- c. Fundamental of Computers, by Balagurusamy E., New Delhi: Tata Mcgraw Hill.

7. Evaluation System

Student academic performance is assessed through internal and external methods. Internal evaluation is carried out by faculty members, monitoring day-to-day student progress. On the other hand, external evaluation is conducted by the university, involving semester-end examinations worth 100 marks. This evaluation system operates as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theoretical Subjects		50	Semester End	50
Daily attendance (Regularity and Punctuality)	10%			
Assignments	20%			
Discipline (Dress, Behavior, outlooks)	10%			
VIVA and Class Performance	10%			
Presentation skill	10%			